

Chemical Handling and Storage SOP

Document No: SOP-CHM-001 | Category: Chemical | Revision: 3 | Effective Date: 2026-01-15 | Review Cycle: Annual

1. Purpose

This SOP establishes safe practices for the receipt, storage, handling, use, and disposal of hazardous chemicals within the facility to protect employees, prevent environmental contamination, and ensure compliance with applicable chemical safety regulations.

2. Scope

Applies to all employees and contractors involved in the procurement, storage, transfer, mixing, application, or disposal of chemicals including acids, alkalis, solvents, paints, adhesives, and cleaning agents used across the facility.

3. Safety Data Sheets (SDS)

A current Safety Data Sheet must be obtained and reviewed for every chemical before it is brought on site. SDS binders are maintained in the chemical storage area, the production floor, and the site safety office, and are also available electronically to all employees. Employees must review the relevant SDS before first use of any chemical, paying particular attention to sections on hazards, exposure limits, required PPE, and first-aid measures.

4. Labeling Requirements

- All chemical containers must retain the manufacturer's original label showing product identity, hazard pictograms, and precautionary statements as per the Globally Harmonized System (GHS).
- Secondary containers used for decanting must be labeled with the chemical name, hazard class, and date of transfer before leaving the original container's presence.
- Unlabeled containers of unknown chemicals must never be used; they must be quarantined and reported to the EHS department for identification and safe disposal.

5. Storage Requirements

- Chemicals must be segregated by compatibility class — acids separated from bases, oxidizers separated from flammables — per the facility's chemical segregation chart.
- Flammable liquids must be stored in approved flammable storage cabinets away from ignition sources, with quantities limited per local fire code.
- Storage areas must have adequate ventilation, secondary spill containment (bundling), and be located away from drains connected to storm water systems.
- Temperature-sensitive chemicals must be stored within the range specified on the SDS, with temperature monitoring where required.

6. Personal Protective Equipment

Task-specific PPE, as identified in the SDS and the area risk assessment, must be worn during all chemical handling, typically including chemical-resistant gloves, splash goggles or face shields,

and chemical-resistant aprons or coveralls. Respiratory protection is required when handling volatile chemicals in areas without adequate local exhaust ventilation, and must be selected per the facility's respiratory protection program.

7. Handling and Transfer Procedures

- Always add acid to water, never water to acid, when diluting concentrated acids, to prevent violent exothermic splashing.
- Use appropriate transfer equipment such as pumps, funnels, or drum dispensers rather than manual pouring wherever possible to reduce splash risk.
- Never mix chemicals unless the mixing procedure is explicitly documented and approved; unintentional mixing (e.g., bleach and ammonia-based cleaners) can generate toxic gases.
- Work with volatile or strong-smelling chemicals only within designated fume hoods or ventilated areas.

8. Spill Response

Minor spills within an employee's training and the capacity of the on-site spill kit may be contained and cleaned up immediately using absorbent materials appropriate to the chemical class, followed by proper disposal in designated hazardous waste containers. Major spills, spills of unknown chemicals, or spills producing hazardous vapors require immediate area evacuation, activation of the spill response team, and notification of the EHS department; affected floor drains must be blocked to prevent environmental discharge.

9. Waste Disposal and Training

All chemical waste must be collected in properly labeled hazardous waste containers and disposed of through the facility's licensed hazardous waste contractor; chemical waste must never be poured down drains or disposed of with general waste. All employees handling chemicals receive GHS and chemical safety training during induction, with annual refreshers and additional training whenever a new chemical is introduced to their work area.